A

1. Option (d)

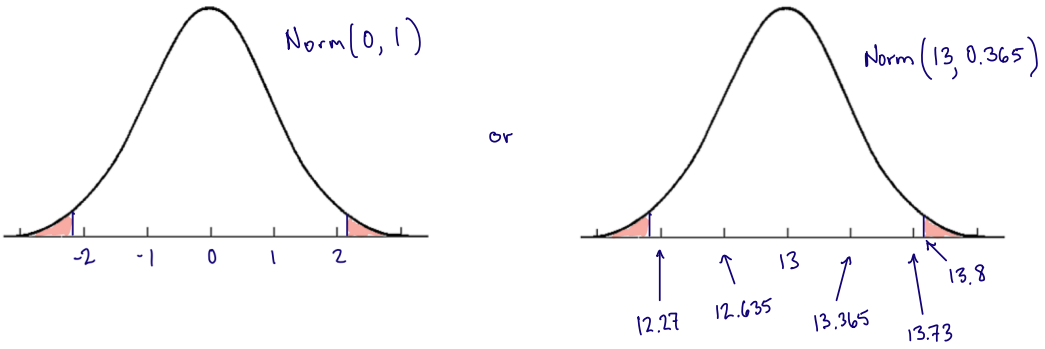
2. (a) variable: How old when began smoking?

   (b) $H_0: \mu = 13.0$ $\quad H_a: \mu \neq 13.0$

   (c) $SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{2.1}{\sqrt{33}} = 0.36556$

   (d) $Z = \frac{13.8 - 13}{0.36556} = 2.1884$

   (e)



3. Type II error, in this instance would mean that the actual portion of adult males in this age range attending church was different than 14%, but you did not reject the proposal (H_0) that it equals 14%.